ENEL 573 Midterm Study Guide

Built from: lecture slides (Ch. 1, 2, 3), HW Assignments 1 to 3, the Sample Midterm, the Practice Questions, the Project 1 and 2 manuals, and Kurose & Ross, *Computer Networking: A Top-Down Approach* (9th ed.).

> **Exam facts (from the course announcements)**
>
> - 50 minutes, closed book, **3 written-response problems**, each with several sub-parts.
> - Non-programmable scientific calculator allowed. Bring your UCID card and a pen.
> - Coverage: **Ch. 1** (network edge, network core, packet vs circuit switching), **Ch. 2** (network analysis and queueing theory), **Ch. 3 up to the end of 3.5** (error detection/correction, retransmission, MAC protocols). Link-layer addressing/ARP (3.6) is **not** on the midterm.
> - Formulas: you are **not** expected to memorize derivations. You **must** know the M/M/1 results and anything at the homework level. Any "complex" formula (for example M/M/1/N blocking) will be given on the exam.

---

0. What the exam actually looks like

The sample midterm was built almost entirely from problems you have already seen. Use that as your main hint.

Sample midterm problem	Topic	Where it came from
P1 (a to d)	Circuit vs packet switching, binomial probability	Textbook Ch. 1 **P8**, HW1 **P4**, slide 1-36
P2 (1 to 3)	M/M/1 minimum service rate	**Identical** to HW2 **P3** and lecture slide 2-47
P3.1	CSMA/CD backoff wait time (K × 512 bit times)	Practice **P2(b)**, textbook Ch. 6 **P17**
P3.2	CRC check at the receiver	Practice **P3(a)**

Takeaway: if you can redo every HW problem (HW1 to HW3), every Practice Question, and the sample midterm without notes, you are in very good shape. Section 6 lists the textbook problems that match.

Tonight's priority checklist (most points per hour first)

1. ☐ M/M/1 formulas (Section 2.4): write them from memory, then redo HW2 P2 and P3.
2. ☐ CRC long division (sender **and** receiver) and 2D parity (HW2 P4).

3. ☐ Delay problems: store-and-forward, message vs packet switching, multi-hop pipelining (HW1 P1, P2, P5; HW2 P1; Practice P4).
4. ☐ ALOHA throughput formulas and the HW3 probability questions; CSMA/CD backoff and collision-detection time.
5. ☐ Stop-and-wait utilization and pipelining window (Practice P1 a to c).
6. ☐ Circuit vs packet switching binomial question (Sample P1).
7. ☐ Concept short answers (Section 7) once you are tired of computing.

---

1. Chapter 1: Introduction

1.1 What is the Internet?

- **"Nuts and bolts" view:** billions of connected **hosts/end systems**, **communication links** (fiber, copper, radio, satellite) with a **bandwidth** (transmission rate, bps), and **packet switches** (routers, link-layer switches, base stations/APs) that forward packets. The Internet is a **network of networks**, loosely hierarchical.
- **Service view:** an infrastructure that provides services to distributed applications (Web, VoIP, email, games). It offers reliable delivery or "best effort" (unreliable) delivery.
- **Protocol:** defines the **format** and **order** of messages exchanged between entities, and the **actions** taken on transmission/receipt (e.g., TCP, IP, HTTP, Ethernet). Standards: RFCs from the IETF.

1.2 Network edge

- **End systems (hosts)** run applications at the edge.
  - **Client/server:** client requests service from an always-on server (Web, email).
  - **Peer-to-peer:** minimal or no dedicated servers.
- **Access networks** connect end systems to the first (edge) router. Always ask: *what bandwidth? shared or dedicated?*

Access technology	Key facts
Dial-up modem	Existing phone line, up to 56 kbps, cannot phone and surf at the same time, not “always on”
DSL	Existing phone line to the central office (**dedicated** line), DSLAM; frequency split: 0 to 4 kHz voice, 4 to 50 kHz upstream, 50 kHz to 1 MHz downstream; asymmetric
Cable (HFC)	Cable TV infrastructure, hybrid fiber coax, asymmetric; homes **share** the link to the router (unlike DSL)
Ethernet	Companies/universities; 10 Mbps to 100 Gbps; shared medium uses **CSMA/CD**
Wireless	WLAN (802.11 WiFi) via access point; wide-area cellular (LTE, NB-IoT, LoRaWAN)

- **Physical media:** *guided* (twisted pair, coax, fiber) vs *unguided* (radio).
  - Twisted pair: Cat 3 (10 Mbps), Cat 5 (100 Mbps), Cat 5e (1 Gbps), Cat 6 (10 Gbps).
  - Coax: baseband (single channel) or broadband (multiple channels, HFC).
  - Fiber: light pulses, very high speed, **low error rate**, immune to EM noise, repeaters far apart.
  - Radio: no wire, affected by reflection, obstruction, interference. Types: terrestrial microwave, WiFi LAN, cellular, satellite (10 to 100 Mbps, long end-to-end delay; geosynchronous vs low-altitude).

1.3 Network core: packet switching vs circuit switching

Packet switching

- Message is broken into **packets** (header + payload + trailer). Header: source/destination addresses, sequence number. Trailer: error-check bits.
- Each packet uses the **full link rate**; no reservation. Resources used on demand → **statistical multiplexing** (no fixed timing pattern).
- **Store-and-forward:** a node must receive the *entire* packet before it can transmit it on the next link.
- Congestion → packets **queue** (delay) and may be **dropped** (loss) when the buffer is full.
- **Pipelining gain:** splitting a message into small packets lets different links work in parallel, reducing total delivery time.

Circuit switching

- End-to-end resources (link bandwidth, switch capacity) **reserved** for the call; **call setup** required; dedicated, no sharing (idle pieces are wasted).
- Link divided into “pieces” by **FDM** (frequency bands) or **TDM** (time slots in a revolving frame).

- **“Circuit-like” (guaranteed) performance:** constant data rate and constant end-to-end delay because resources are reserved along the path.
- No store-and-forward: bits stream through the switches.

Item	Circuit switching	Packet switching
Dedicated path	Yes	No
Bandwidth available	Fixed	Dynamic
Potentially wasted bandwidth	Yes	No
Store-and-forward	No	Yes
Call setup	Yes	Not needed
When can congestion occur	At setup time (call blocked)	On every packet
Charging	Per minute	Per packet
Each packet follows the same route	Yes	No

> **Why packet switching supports more users (slide 1-36):** 1 Mbps link, each user needs 100 kbps when active, active 10% of the time. Circuit switching: 10 users. Packet switching with 35 users: P(more than 10 active) ≈ 0.0004. Great for **bursty** data. Circuit switching is better for **steady, long-lived** traffic (e.g., constant-rate streaming, telephone) that needs guaranteed rate/delay.

Recognizing multiplexing from a capture (Practice P1d): a strict repeating pattern of owners (A, B, A, B with fixed positions, empty slots when idle) suggests **TDM**; irregular order (e.g., “A A A” back-to-back) means **statistical multiplexing**.

Internet structure: tier-1 ISPs (global, peer with each other for free) and large content providers (Google, Akamai) at the center; tier-2 (regional) ISPs pay tier-1 providers and may peer at **IXPs**; tier-3/local ISPs are the access networks.

1.4 The way networks work: layering

- **Layering** gives modularity: each layer provides a service to the layer above using the service of the layer below. Layers can be changed independently; heterogeneous systems can interoperate (“open systems”).
- **Protocol stack** = set of layers and protocols. **Peer-to-peer protocol** = layer N at source talks to layer N at destination (conceptually) using the lower layers. **Interface** = services a layer provides to the layer above.

OSI layer	Function
7 Application	Common services: file transfer, directory, virtual terminal
6 Presentation	Compression, security (encryption), format conversion
5 Session	Set up and supervise connections between end systems
4 Transport	End-to-end delivery; may retransmit erroneous packets
3 Network	Route packets from source to destination across many links
2 Data link	Reliable transfer between nodes on the **same physical link**
1 Physical	Raw bit stream over the channel

- Layers 1 to 3: network support. Layers 5 to 7: user support. Layer 4 links the two.
- **TCP/IP 5-layer stack:** application (HTTP, SMTP, FTP), transport (TCP, UDP), network (IP, routing), link (Ethernet, WiFi MAC), physical.
- **Encapsulation names:** message (application) → **segment** (transport, adds H_t) → **datagram** (network, adds H_n) → **frame** (link, adds H_l and trailer).
- Switches are layer-2 devices (examine link headers); routers are layer-3 devices (examine network headers). Both are store-and-forward.

1.5 Performance: delay, loss, throughput

$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

Delay	Formula / meaning
Processing d_{proc}	Check bit errors, choose output link; typically under a millisecond
Queueing d_{queue}	Waiting for the output link; depends on congestion
Transmission d_{trans}	L/R: time to **push** all L bits onto a link of rate R
Propagation d_{prop}	d/s: time for one bit to **travel** a link of length d at speed s (about 2 to 3×10^8 m/s)

- **Caravan analogy:** toll booth service time = transmission delay; driving time between booths = propagation delay. The first bit can reach the next router before the last bit has left the current one (if $d_{prop} < d_{trans}$).
- **Traffic intensity** La/R (a = average packet arrival rate in packets/s). $La/R \approx 0$: small queueing delay. $La/R \to 1$: delay grows very large. $La/R > 1$: queue grows without bound. (This is exactly $\rho = \lambda/\mu$ from Chapter 2.)

- **Loss:** a packet arriving at a full buffer is dropped; may be retransmitted by the previous node, the source, or not at all.
- **Throughput:** rate (bits/s) at which bits are delivered. End-to-end throughput = rate of the **bottleneck link**: $\min(R_1, R_2, \ldots)$. With M connections fairly sharing a backbone link R: $\min(R_s, R_c, R/M)$.
- **Traceroute:** sends 3 probes to each router on the path and measures round-trip times; `*` = no reply.

> **Key Chapter 1 formulas**
>
> - One packet, N store-and-forward links of rate R: $d_{end\text{-}end} = N\frac{L}{R}$ (plus $\sum d_{prop}$, $\sum d_{proc}$)
> - P packets back-to-back over N equal links: $(N + P - 1)\frac{L}{R}$ (+ propagation)
> - P packets over links with different rates: $\sum_i \frac{L}{R_i} + (P-1)\frac{L}{R_{min}}$
> - Message switching (whole file F as one message) over N links: $N\frac{F}{R}$
> - Circuit switching with k FDM/TDM channels: rate per call R/k; time $= t_{setup} + \frac{F}{R/k} + d_{prop}$
> - Circuit-switched users: $N = \lfloor R/r_{user} \rfloor$. Packet switching with M users, each active w.p. p: $P(\text{exactly } n \text{ active}) = \binom{M}{n} p^n (1-p)^{M-n}$, $P(\text{more than } N) = 1 - \sum_{k=0}^{N} \binom{M}{k} p^k (1-p)^{M-k}$
> - Bandwidth-delay product $R \cdot d_{prop}$ = max bits "in flight" on the link; width of one bit $= s/R$ meters.

1.6 Chapter 1 worked examples

E1. HW1 P2 (30 Mbit MP3, 10 Mbps links, 10,000 km, $s = 2 \times 10^8$ m/s).

1. $d_{trans} = 30 \times 10^6 / 10 \times 10^6 = 3$ s.
2. $d_{prop} = 10^7 / (2 \times 10^8) = 0.05$ s, so end-to-end $= 3.05$ s.
3. Bits sent when the first bit arrives $= R \cdot d_{prop} = 10^7 \times 0.05 = 500{,}000$ bits.
4. Two links (5,000 km each), one store-and-forward router: $2(3) + 2(0.025) = 6.05$ s.
5. Three 10-Mbit packets, 2 links: $(3 + 2 - 1)(1\text{ s}) + 0.05 = 4.05$ s.
6. 10 FDM channels → 1 Mbps per channel: $30 + 0.05 = 30.05$ s.

E2. HW2 P1 (message segmentation, 3 links of 1.5 Mbps, 7.5 Mbit message).

- No segmentation: $7.5 \times 10^6 / 1.5 \times 10^6 = 5$ s per hop, total $3 \times 5 = 15$ s.
- 5,000 packets of 1,500 bits: 1 ms per packet per hop. First packet reaches switch 1 at 1 ms; second packet fully at switch 1 at 2 ms. Whole file: $(5000 + 3 - 1)(1\text{ ms}) = 5.002$ s, about 3 times faster.
- Drawbacks of segmentation: header overhead per packet, packets must be sequenced and reassembled, more processing, more chance one packet is lost.

E3. HW1 P1 (two nodes between A and B ⇒ 3 links, $M = 15$ MB, $C = 12$ Mbps).

- Message switching: $3 \times \frac{8M}{C} = 3 \times \frac{120 \times 10^6}{12 \times 10^6} = 30$ s.
- Packet switching with $L = 9600$ bits: $P = 120 \times 10^6 / 9600 = 12,500$ packets; $(P + 2)\frac{L}{C} = 12,502 \times 0.8$ ms $= 10.0016$ s.

E4. HW1 P3 / textbook P6 (where is the bit?). At $t = d_{trans}$ the last bit has just left Host A. If $d_{prop} > d_{trans}$, the first bit is in the link, $s \cdot d_{trans}$ meters from A. If $d_{prop} < d_{trans}$, the first bit has already reached B. Setting $d_{prop} = d_{trans}$: $m = Ls/R$ (e.g., $L = 1000$, $R = 284$ kbps, $s = 2.5 \times 10^8$ gives $m \approx 880$ km).

E5. HW1 P5 (two links, $d_{proc} = 1$ ms, 1,000 bytes, 1 Mbps, 4,000 km + 1,000 km, $s = 2.5 \times 10^8$). $d = \frac{L}{R_1} + \frac{d_1}{s_1} + d_{proc} + \frac{L}{R_2} + \frac{d_2}{s_2} = 8 + 16 + 1 + 8 + 4 = 37$ ms.

E6. Practice P4 (broadcast through switches, 100 KB file, 1 KB packets). A to S1 = 10 Mbps, S1 to S2 = 10 Mbps, S1 to D = 1 Mbps, S2 to C = 2 Mbps, S2 to B = 4 Mbps. One 1 KB packet takes 0.8 ms on 10 Mbps, 2 ms on 4 Mbps, 4 ms on 2 Mbps, 8 ms on 1 Mbps.

- First packet at D: $0.8 + 8 = 8.8$ ms. At C: $0.8 + 0.8 + 4 = 5.6$ ms. At B: $0.8 + 0.8 + 2 = 3.6$ ms.
- Whole file at C: $5.6 + 99 \times 4 = 401.6$ ms. At B: $3.6 + 99 \times 2 = 201.6$ ms.
- Throughput: C ≈ 249 KB/s, B ≈ 496 KB/s. B is better because its bottleneck (4 Mbps) is faster than C's (2 Mbps).
- **Trap:** after the first packet, packets arrive every L/R_{min}, where R_{min} is the slowest link on the path, which is not always the last link.

E7. Sample P1 (2 Mbps link, users need 300 kbps, active 12%). (a) $\lfloor 2000/300 \rfloor = 6$ users. (b) $p = 0.12$. (c) $\binom{M}{n}(0.12)^n(0.88)^{M-n}$. (d) $P(\geq 7) = 1 - \sum_{k=0}^{6} \binom{M}{k}(0.12)^k(0.88)^{M-k}$.

---

2. Chapter 2: Network Analysis and Queueing Theory

2.1 Why model networks?

- "Analyze first, build later": a model is a mathematical abstraction that keeps only the relevant details.
- Uses: **evaluate** performance (average waiting time, throughput, loss probability from buffer overflow) and **improve** it (choose a service rate for a tolerable delay, size a buffer for a target loss probability).
- **Randomness at the source:** random data generation (web browsing), voice on/off talk spurts, variable video frame sizes. **Randomness in the network:** changing wireless channels, congestion (statistical multiplexing), contention among users. **Buffers** absorb this randomness.

2.2 Probability review

- $P(E) = \lim_{n \to \infty} n(E)/n$ (relative frequency).

- Conditional: $P(E \mid F) = \dfrac{P(E \cap F)}{P(F)}$. Joint: $P(E \cap F) = P(E \mid F)P(F)$. Independent: $P(A \cap B) = P(A)P(B)$.
- **Total probability** (for a partition $E_1, \ldots, E_n$): $P(A) = \sum_i P(A \mid E_i)P(E_i)$.
- CDF $F(x) = P(X \le x)$; pdf $f(x) = dF/dx$; pmf $p(x_i) = P(X = x_i)$.
- $E[X] = \sum x_i p(x_i)$ or $\int x f(x) dx$; $\mathrm{Var}(X) = E[X^2] - (E[X])^2$.

Distribution	pmf / pdf	Mean	Variance	Network use
Binomial(n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$	# active users; # bit errors in a packet
Geometric(p)	$(1-p)^{k-1} p,\ k \ge 1$	$1/p$	$(1-p)/p^2$	# transmissions until first success
Poisson(λ)	$\dfrac{\lambda^k e^{-\lambda}}{k!}$	λ	λ	# arrivals in a unit interval
Exponential(λ)	$\lambda e^{-\lambda x},\ x \ge 0$; CDF $1 - e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	inter-arrival and service times

Examples from the slides

- Packet of n bits, bit error probability p_B: $P(\text{no error}) = (1-p_B)^n$; $P(\text{exactly one}) = np_B(1-p_B)^{n-1}$; $P(\text{packet corrupted}) = 1 - (1-p_B)^n$.
- Stop-and-wait, loss probability p_E: $P(\text{need exactly 3 transmissions}) = p_E^2(1-p_E)$; average transmissions $= \dfrac{1}{1-p_E}$.
- **Memoryless property** (exponential only): $P(X > x + t \mid X > t) = P(X > x)$.
- Bank example (time in bank exponential, mean 10 min, $\mu = 1/10$): $P(X > 15) = e^{-1.5} \approx 0.223$. Given still there after 10 min, $P(X > 15 \mid X > 10) = P(X > 5) = e^{-0.5} \approx 0.607$.
- **Poisson process** with rate λ: number of arrivals in an interval of length t is Poisson with mean λt: $P(X(t) = k) = \dfrac{e^{-\lambda t}(\lambda t)^k}{k!}$. Inter-arrival times are i.i.d. exponential(λ), because $P(\tau \le s) = 1 - P(\text{no arrival in } s) = 1 - e^{-\lambda s}$.
- **Generating exponential samples (Project 1):** inverse transform. If $U \sim U(0,1)$, then $X = -\dfrac{1}{\lambda}\ln(1-U)$ (equivalently $-\ln U/\lambda$) is exponential(λ).

2.3 Queueing basics

- Single-server queue: arrivals → buffer → server (the transmission link). Service time is proportional to packet length.
- Multi-server: an arrival goes to any empty server.
- **Kendall notation A/S/C/K:** A = arrival process, S = service time distribution, C = number of servers, K = max number in the system (buffer + servers, infinite if omitted). **M** = Markovian (Poisson arrivals / exponential times, memoryless). G = general.
  - M/M/1: Poisson arrivals, exponential service, 1 server, infinite buffer.
  - M/M/1/N: same, at most N in the system. M/M/m: m servers. M/M/m/m: m servers, no waiting room. G/G/1: general.

2.4 The M/M/1 queue (KNOW THIS COLD)

Setup. Arrival rate λ (Poisson), service rate μ (exponential, mean service time $1/\mu$), $\rho = \lambda/\mu < 1$ for stability.

Derivation outline. Birth-death chain on states $0, 1, 2, \ldots$ (λ up, μ down). Balance (flow in = flow out): $\lambda P_0 = \mu P_1$ and $(\lambda + \mu)P_n = \lambda P_{n-1} + \mu P_{n+1}$ for $n \geq 1$. This gives $P_n = \rho^n P_0$; normalizing $\sum P_n = 1$ gives $P_0 = 1 - \rho$.

Quantity	Result
Utilization (P server busy)	$\rho = \lambda/\mu$
P(system empty / server idle)	$P_0 = 1 - \rho$
P(exactly n in system)	$P_n = (1-\rho)\rho^n$
P(at least n in system)	$P(N \geq n) = \rho^n$
Avg number in system	$L = E[N] = \frac{\rho}{1-\rho} = \frac{\lambda}{\mu - \lambda}$
Avg number waiting in queue	$E[N_q] = \frac{\rho^2}{1-\rho} = \frac{\lambda^2}{\mu(\mu-\lambda)}$
Avg time in system	$W = \frac{1}{\mu - \lambda}$
Avg waiting time in queue	$T_q = W_q = \frac{\rho}{\mu - \lambda} = \frac{\lambda}{\mu(\mu-\lambda)}$
Relationship	$W = T_q + \frac{1}{\mu}$

Little's Law (any stable system): $E[N] = \lambda W$ and $E[N_q] = \lambda T_q$.

- **Units:** if packets have mean length $\bar{L}$ bits and the link is C bps, then $\mu = C/\bar{L}$ packets/s. Keep λ and μ in the **same** units.
- **"Queue" wording trap:** "at least 20 people **in the queue**" (waiting) means at least 21 **in the system**: $P(N \geq 21) = \rho^{21}$. "Queue length" in the sample solution means $E[N_q]$.
- **Example 2.2 (scaling):** multiply both λ and μ by k: $L = \frac{k\lambda}{k\mu - k\lambda} = \frac{\lambda}{\mu - \lambda}$ is unchanged, but $W = \frac{1}{k(\mu - \lambda)}$ is k times smaller. This is why **statistical multiplexing beats TDM/FDM**: one fast shared queue beats k slow dedicated queues.

2.5 M/M/1/N (finite buffer): formula likely provided on the exam

- At most N packets in the system; arrivals that find the system full are **blocked (dropped)**.
- $P_n = \frac{1-\rho}{1-\rho^{N+1}}\rho^n$ for $0 \leq n \leq N$ (valid for any $\rho \neq 1$, even $\rho > 1$). If $\rho = 1$: $P_n = \frac{1}{N+1}$.
- **Blocking probability** $P_B = P_N = \frac{(1-\rho)\rho^N}{1-\rho^{N+1}}$.
- **Throughput** $= \lambda(1 - P_B) = \mu(1 - P_0)$.

2.6 Networks of M/M/1 queues

- Each router i is an M/M/1 queue with total arrival rate λ_i = sum of all flows γ_j passing through it (assume independence).
- $E[N_i] = \dfrac{\lambda_i}{\mu_i - \lambda_i}$; total $E[N] = \sum_i \dfrac{\lambda_i}{\mu_i - \lambda_i}$.
- Average end-to-end delay (Little): $E[T] = \dfrac{E[N]}{\lambda}$, where $\lambda = \sum_j \gamma_j$ is the total **external** arrival rate.
- Example: $\gamma_1 = 10$, $\gamma_2 = 5$, $\gamma_3 = 5$ packets/s, all $\mu_i = 30$; $\lambda_1 = \gamma_1 + \gamma_2 = 15$, $\lambda_2 = \gamma_1 + \gamma_2 + \gamma_3 = 20$, $\lambda_3 = \gamma_1 + \gamma_3 = 15$. $E[N] = 1 + 2 + 1 = 4$, $E[T] = 4/20 = 0.2$ s.

2.7 Chapter 2 worked examples

E8. HW2 P3 = Sample P2 ($\lambda = 15$/hr, find minimum μ).

1. Idle at least 10%: $P_0 = 1 - \rho \geq 0.1 \Rightarrow \lambda/\mu \leq 0.9 \Rightarrow \mu \geq 15/0.9 = 16.67$/hr.
2. $E[N_q] = \dfrac{\lambda^2}{\mu(\mu - \lambda)} \leq 10 \Rightarrow 225 \leq 10\mu^2 - 150\mu \Rightarrow \mu^2 - 15\mu - 22.5 \geq 0 \Rightarrow \mu \geq \dfrac{15 + \sqrt{315}}{2} = 16.374$/hr (slides round to 16.375).
3. At least 20 in queue = at least 21 in system: $\rho^{21} \leq 0.5 \Rightarrow \rho \leq 0.5^{1/21} = e^{\ln(0.5)/21} = 0.96753 \Rightarrow \mu \geq 15/0.96753 = 15.50$/hr.
4. Overall minimum to satisfy **all** three: the largest, $\mu \geq 16.67$/hr.

E9. HW2 P2 (50 kbps line, 10 sessions, 150 packets/min each, mean 1000 bits). Per session $\lambda = 2.5$ packets/s.

- i. **TDM**, 10 channels of 5 kbps: $\mu = 5$ packets/s, $\rho = 0.5$. $E[N_q] = 0.5$, $E[N] = 1$, $W = 1/(5 - 2.5) = 0.4$ s per session.
- ii. **Statistical multiplexing**, one 50 kbps queue: $\lambda = 25$, $\mu = 50$, $\rho = 0.5$. Whole system: $E[N_q] = 0.5$, $E[N] = 1$, $W = 1/25 = 0.04$ s. Per session (by Little): $E[N] = 2.5 \times 0.04 = 0.1$, $E[N_q] = 0.05$.
- Same ρ, but delay is **10 times smaller** with statistical multiplexing (Example 2.2 with $k = 10$).

E10. M/M/1/N. $\rho = 0.5$, $N = 3$: $P_B = \dfrac{0.5 \times 0.125}{1 - 0.0625} = 0.0667$; throughput $= 0.933\lambda$.

---

3. Chapter 3: Data Link Layer (3.1 to 3.5)

3.1 Services

- Nodes = hosts and routers; links = channels between adjacent nodes; layer-2 packet = **frame** (encapsulates a datagram).
- Services: **framing**, **link access** (MAC for shared media), **reliable delivery** between adjacent nodes (mostly on high-error wireless links; rarely on fiber), **flow control**, **error detection**, **error correction**.

- Why both link-level and end-to-end reliability? Link-level recovery is faster/local on noisy links; end-to-end is still needed for losses inside routers/the network.
- Implemented in the **network adapter (NIC)**: hardware + software + firmware. Sender encapsulates and adds error-check bits; receiver checks errors, extracts the datagram.

3.2 Framing

- Why frames: big blocks have a higher error probability, only small pieces need retransmission, and small frames stop one node hogging a shared medium.
- **Fixed-size** framing: hard to resynchronize after bit loss. **Variable-size**: byte count or flags.
- **Byte count:** header field gives the frame length; if the count is corrupted the receiver loses sync.
- **Flag + bit stuffing:** frames start and end with flag `01111110`. Sender inserts a `0` after every five consecutive `1`s in the data. Receiver: after five `1`s, next bit `0` → remove it (stuffed); next bits `10` → flag; next bits `11` → error.
  - Example: data `0111111111100` → stuffed `011111011111000` (a 0 inserted after each run of five 1s).

3.3 Error detection and correction

- Errors from noise/interference; BER (e.g., 10^{-6}). **Single-bit** vs **burst** errors (two or more bits, not necessarily consecutive). Bursts are more likely: bits affected = rate × noise duration (1/100 s of noise hits 10 bits at 1 kbps, 10,000 bits at 1 Mbps).
- **Detection** (low overhead, retransmit on error; fine on fiber/copper) vs **correction** (FEC, more overhead; used on noisy wireless links). More redundancy = better detection but more overhead. Detection is never 100% reliable.
- Frame of m data bits + r redundant bits = n bits.

Single parity check

- **Even parity:** parity bit makes the total number of 1s even. **Odd parity:** makes it odd.
- Detects **all odd numbers** of bit errors (so all single-bit errors); misses all even numbers. Overhead $1/(k+1)$.
- For N_b bits with bit error probability P_b: $P_1 = (1-P_b)^{N_b}$ (no error); $P_2 = \sum_{i=1}^{\lfloor N_b/2 \rfloor} \binom{N_b}{2i} P_b^{2i} (1-P_b)^{N_b-2i}$ (undetected); $P_3 = 1 - P_1 - P_2$ (detected).
- Frame of n bytes, each with its own parity bit: $P_{f1} = P_1^n$; $P_{f2} = \sum_{i=1}^{n} \binom{n}{i} P_2^i (1-P_2)^{n-i}$; $P_{f3} = 1 - P_{f1} - P_{f2}$.

Two-dimensional parity

- Arrange data as n rows × m columns; add one parity bit per row, one per column, and one corner bit: $mn + m + n + 1$ total bits.
- **Corrects any single-bit error** (the failing row and column intersect at the bad bit). **Detects** any 2 or 3 errors. Some 4-bit patterns (rectangle corners) are **undetectable**.

E11. HW2 P4.

- `11010111` has six 1s: even parity bit **0**, odd parity bit **1**. `10001010` has three 1s: even parity bit **1**, odd parity bit **0**.
- 2D even parity of `11010010 11100011 00111011` (3 × 8):

Row	Data	Row parity
1	1 1 0 1 0 0 1 0	0
2	1 1 1 0 0 0 1 1	1
3	0 0 1 1 1 0 1 1	1
Column parity	0 0 0 0 1 0 1 0	**0** (corner)

- Received `11010010 11110011 00111011`: row 2 parity fails (six 1s, expected odd) and column 4 parity fails (three 1s). The single error is at **row 2, column 4**; flip it back to 0. 2D parity detects and corrects this single-bit error.

Cyclic Redundancy Check (CRC)

- Generator G has $r+1$ bits (degree r, first and last bit 1); both sides know G.
- **Sender:** append r zeros to D, divide by G with **modulo-2 (XOR, no carries/borrows)** division; the r-bit remainder is R. Send $\langle D,R\rangle$, which is exactly divisible by G.
- **Receiver:** divide the whole received string by G. Remainder all zeros → accept; nonzero → **error detected**.
- Division rules: if the leading bit of the current window is 1, XOR with G; if 0, XOR with zeros (i.e., just shift). Drop the leading bit each step.
- Properties: detects **all single-bit errors**, **all burst errors of length** $\leq r$, all double-bit errors if G is chosen well, all odd numbers of errors if G has factor $(x+1)$. Undetectable: error polynomials $E(x)$ that are multiples of $G(x)$. CRC-16 detects 100% of bursts up to 16 bits, 99.997% of 17-bit bursts. Easy in hardware (shift register + XOR).
- Polynomial view: `110001` $= x^5 + x^4 + 1$.

E12. CRC sender (slide example): $D = 101110$, $G = 1001$ ($r = 3$), so divide `101110` + `000` = `101110000` by `1001`. Work 4 bits at a time:

Step	Current 4-bit window	Leading bit	XOR with	Result	Keep last 3 bits, bring down next bit
1	1011	1	1001	0010	010 + **1** = 0101
2	0101	0	0000	0101	101 + **0** = 1010
3	1010	1	1001	0011	011 + **0** = 0110
4	0110	0	0000	0110	110 + **0** = 1100
5	1100	1	1001	0101	101 + **0** = 1010
6	1010	1	1001	0011	no bits left: R = **011**

Transmitted: `101110 011`. (Slide example 2: $D = 1100$, $G = 1011$ gives $R = 010$.)

E13. CRC receiver (Sample P3.2): received `1010110`, $G = 10011$. Dividing gives remainder `1001` $\neq 0$, so the frame was **not** received correctly.

3.4 Retransmission (ARQ) and flow control

ARQ goal: deliver data **without errors, in order, without duplicates**. Tools: error detection, **ACKs**, **timeouts**, **sequence numbers**.

Stop-and-wait (= Alternating Bit Protocol, ABP; Project 2)

- Send one frame, wait for ACK. No ACK before timeout → resend. Timeout must exceed the round-trip time.
- Lost **ACK** → sender resends → receiver would get a **duplicate**, so frames need sequence numbers (1 bit, mod 2).
- **Premature timeout / delayed ACK** → ACKs must be numbered too (1 bit). ACK carries the **next expected** sequence number.
- **Piggybacking:** in two-way traffic put the ACK in the header of a data frame to save overhead.

Stop-and-wait throughput ($t_l = L/C$ data frame time, τ = one-way propagation, t_A = ACK time)

$$t_T = t_l + 2\tau + t_A, \qquad \lambda_{max} = \frac{1}{t_T} \text{ frames/s}, \qquad U = \frac{t_l}{t_T}$$

- With frame **or** ACK error probability p (timeout = t_T): the number of attempts is geometric, average time per success $t_v = \frac{t_T}{1-p}$, so $\lambda_{max} = \frac{1-p}{t_T}$. Average transmissions $= \frac{1}{1-p}$.

- Slide examples (1500-byte data and ACK, 1 Mbps, $t_l = t_A = 12$ ms): land line 100 km → $t_T = 25$ ms, $U = 48\%$, 0.48 Mbps. Satellite 71,600 km → $t_T = 503.4$ ms, $U = 2.4\%$, 24 kbps. Stop-and-wait wastes long or fast links.

Pipelining (sliding window)

- Allow up to W un-ACKed frames in flight. To keep the link 100% busy: $W \geq \frac{t_T}{t_l} = \frac{t_l + 2\tau + t_A}{t_l}$. Otherwise $U = \frac{Wt_l}{t_T}$.
- **Go-Back-N:** sender window up to $2^m - 1$ (m sequence bits); one timer (oldest un-ACKed); receiver window = 1 (no out-of-order buffering, cumulative ACK of next expected frame); on timeout or full window, **resend all** outstanding frames. Poor on noisy links.
- **Selective Repeat:** receiver window > 1 buffers out-of-order frames; a **NAK** (or individual timer) triggers retransmission of only the missing frame. (Textbook: window $\leq 2^{m-1}$.)
- **Flow control:** receiver slows the sender by delaying ACKs, negotiating window size at setup, or changing window size during the session.

E14. Practice P1 (Earth to Mars, 54.6 million km, 1.5 KB frames, 10 Mbps, $c = 3 \times 10^8$).

- $t_l = 12,000/10^7 = 1.2$ ms. $\tau = 5.46 \times 10^{10}/3 \times 10^8 = 182$ s. Copy one frame ≈ 182.0012 s.
- Max frames before waiting for an ACK: $\frac{t_l + 2\tau}{t_l} = \frac{364.0012}{0.0012} \approx 303,334$ frames.

3.5 Multiple access (MAC) protocols

- **Point-to-point** (PPP) vs **broadcast** (shared Ethernet, WiFi, satellite). Two simultaneous transmissions → **collision** at the receiver. The MAC protocol is a distributed algorithm deciding who transmits; coordination must use the channel itself.
- **Ideal MAC** (rate R): one node sends at R; M nodes average R/M each; fully decentralized (no master, no clock sync); simple.

Class	Examples	Notes
Channel partitioning	TDMA, FDMA, CDMA	Fixed share R/N; idle slots/bands wasted; no collisions; common in cellular
Random access	Pure ALOHA, slotted ALOHA, CSMA, CSMA/CD, CSMA/CA	Full rate R, collisions possible, need detection + random backoff
Taking turns	Polling, token passing	Polling: master invites slaves (overhead, latency, single point of failure). Token: token overhead, latency, token loss

ALOHA analysis ($X = L/R$ = frame time; G = attempts per X (new + retransmissions, Poisson); S = successes per X)

	Pure ALOHA	Slotted ALOHA
Transmit when	Any time	Only at slot starts (needs clock sync)
Vulnerable period	$2X$	X
Throughput	$S = Ge^{-2G}$	$S = Ge^{-G}$
Maximum	$\frac{1}{2e} \approx 0.184$ at $G = 0.5$	$\frac{1}{e} \approx 0.368$ at $G = 1$
Finite N nodes, prob p	$Np(1-p)^{2(N-1)}$	$Np(1-p)^{N-1}$, best $p = 1/N$

- **Why slotted beats pure (Practice P3b):** in pure ALOHA a frame collides with any frame that started within X before **or** after it ($2X$ vulnerable window). Slotting forces starts to slot boundaries, so only frames in the same slot collide (X window).
- Slotted ALOHA with Poisson load G per slot: $P(\text{idle}) = e^{-G}$, $P(\text{success}) = Ge^{-G}$, $P(\text{collision}) = 1 - e^{-G} - Ge^{-G}$.
- Probability a given attempt succeeds $= e^{-G}$ (slotted). Probability of exactly k collisions then a success $= (1-e^{-G})^k e^{-G}$. Expected attempts per success $= G/S = e^{G}$ (pure: e^{2G}).
- For small G, $S \approx G$; for large G, many backlogged users and $S \to 0$.
- **Pure ALOHA receiver (Practice P3c):** wait for a frame; check bit errors; if none, deliver data up and send an ACK; otherwise discard; repeat.

E15. HW3 P1. (a) Each user: 1000 bits / 100 s = 10 bps. Max $= 0.184 \times 56,000 = 10,304$ bps, so $N = 1030$ users. (b) $10,000 \times 18/3600 = 50$ requests/s; slots/s $= 1/125\,\mu s = 8000$; $G = 50/8000 = 0.00625$ requests/slot.

E16. HW3 P2 (G = 50 req/s × 40 ms = 2 per slot). (a) $e^{-2} = 0.135$. (b) $(1-e^{-2})^k e^{-2} = 0.865^k(0.135)$. (c) $e^2 \approx 7.39$ attempts.

E17. HW3 P3 (3 nodes, each transmits w.p. p). One given node succeeds in a slot: $p(1-p)^2$. (a) A's first success in slot 4: $[1-p(1-p)^2]^3\, p(1-p)^2$. (b) Some node succeeds in slot 2: $3p(1-p)^2$. (c) First success (any node) in slot 4: $[1-3p(1-p)^2]^3\, 3p(1-p)^2$.

E18. Practice P5 (Poisson n with mean $\bar{n}$, each sends w.p. $1/\bar{n}$). $P(\text{idle}) = \sum_k (1-1/\bar{n})^k e^{-\bar{n}} \bar{n}^k/k! = e^{-1}$, and $P(\text{success}) = e^{-1}$.

CSMA family

- **CSMA:** listen before talking; if busy wait (or back off), if idle transmit. Collisions still happen because of **propagation delay** (vulnerable period $\approx t_{prop}$). A collision wastes the whole frame time. If $t_{prop} > X$, no gain over ALOHA.
- **CSMA/CD (Ethernet, IEEE 802.3):** (1) wait until idle; (2) transmit and **listen while transmitting**; (3) on collision **abort**, wait a random backoff, go to (1). Wastes only the detection time instead of the whole frame.
  - Worst-case collision detection time $= 2t_{prop}$ (B starts just before A's first bit arrives).
  - **Minimum frame size:** to be sure a collision is seen while still transmitting, $L/R \geq 2t_{prop}$ (Ethernet: 512 bits = 64 bytes).

    - **Backoff:** wait $K \times 512$ bit times, bit time $= 1/R$. Binary exponential backoff (textbook): after the nth collision choose K uniformly from $\{0, 1, \ldots, 2^n - 1\}$ (cap n at 10).
    - Textbook efficiency: $\frac{1}{1 + 5t_{prop}/t_{trans}}$ (goes to 1 as $t_{prop} \to 0$ or frames get long).
- **CSMA/CA (802.11 WiFi):** exchange **RTS / CTS / DATA / ACK**. Short control frames make collisions cheap; RTS/CTS gives "virtual" carrier sense against the **hidden terminal** problem. Less efficient than CSMA/CD. ACKs are needed because a wireless sender cannot detect collisions at the receiver.

E19. Sample P3.1 / Practice P2b: $K = 100$ at 100 Mbps: $100 \times 512 \times 10^{-8} = 512\,\mu$s (at 1 Gbps: 51.2 µs).

E20. Practice P2c: collision detected after 60 bytes (480 bits) on 10 Mbps, longest ever ⇒ $2t_{prop} = 480$ bit times ⇒ $t_{prop} = 240$ bit times $= 240 \times 10^{-7} = 24\,\mu$s.

E21. Practice P2a: 2,500 km fiber at 2.5×10^8 m/s, 2 Mbps, 1000 bytes: $t_{prop} = 10$ ms, $t_{trans} = 4$ ms, total 14 ms.

E22. Practice P1f: most collisions among {pure ALOHA, slotted ALOHA, CSMA/CD}: **pure ALOHA**.

---

4. One-page formula sheet

Topic	Formula
Transmission delay	$d_{trans} = L/R$
Propagation delay	$d_{prop} = d/s$
Nodal delay	$d_{proc} + d_{queue} + d_{trans} + d_{prop}$
P packets, N equal links	$(N + P - 1)L/R + \sum d_{prop}$
Different link rates	$\sum_i L/R_i + (P - 1)L/R_{min}$
Traffic intensity	$La/R = \rho$
Bandwidth-delay product	$R \cdot d_{prop}$
Binomial	$\binom{M}{n} p^n (1-p)^{M-n}$
Poisson	$\lambda^k e^{-\lambda}/k!$, process: $(\lambda t)^k e^{-\lambda t}/k!$
Exponential	$f = \lambda e^{-\lambda x}$, $P(X > x) = e^{-\lambda x}$, mean $1/\lambda$
M/M/1	$P_n = (1-\rho)\rho^n$, $P(N \geq n) = \rho^n$, $L = \rho/(1-\rho)$, $L_q = \rho^2/(1-\rho)$, $W = 1/(\mu - \lambda)$, $W_q = \rho/(\mu - \lambda)$
Little	$L = \lambda W$, $L_q = \lambda W_q$
M/M/1/N	$P_B = (1-\rho)\rho^N/(1-\rho^{N+1})$; throughput $\lambda(1 - P_B)$
Queue network	$E[N] = \sum \lambda_i/(\mu_i - \lambda_i)$, $E[T] = E[N]/\sum \gamma_j$
2D parity bits	$mn + m + n + 1$ total
Stop-and-wait	$t_T = t_l + 2\tau + t_A$, $U = t_l/t_T$, $\lambda_{max} = (1-p)/t_T$
Window to fill pipe	$W \geq t_T/t_l$; GBN window $\leq 2^m - 1$
Pure ALOHA	$S = Ge^{-2G}$, max $1/(2e) = 18.4\%$
Slotted ALOHA	$S = Ge^{-G}$, max $1/e = 36.8\%$; idle e^{-G}; attempts per success e^G
CSMA/CD	detect within $2t_{prop}$; $L_{min} = 2t_{prop}R$; wait $K \cdot 512/R$

5. Common mistakes to avoid

1. **Bits vs bytes.** Multiply bytes by 8. In this course 1 KB = 1000 bytes and 1 Mbps = 10^6 bps (the practice solutions use 1.5 KB = 12 kbits).
2. **Mixing up transmission and propagation delay.** Transmission depends on L and R; propagation depends on distance and speed only.
3. **Forgetting store-and-forward:** each hop adds a full L/R for the first packet. Circuit switching does **not** store-and-forward.
4. **Using the last link as the bottleneck** when a slower link is earlier on the path.

5. **Queue vs system:** "in the queue" excludes the packet in service; $N_q = N - 1$ when busy. Use $E[N_q]$ vs $E[N]$ correctly and translate "at least k in queue" to $N \geq k + 1$.
6. **Unstable queue:** M/M/1 formulas need $\lambda < \mu$. If $\rho \geq 1$ the delay is infinite.
7. **λ and μ in different units** (per hour vs per minute, packets vs bits).
8. **CRC:** append exactly r zeros (one less than the length of G) at the sender; do **not** append zeros at the receiver. Use XOR, never subtraction with borrows. The remainder always has r bits.
9. **Parity:** even parity makes the total count of 1s (data + parity) even.
10. **ALOHA G:** convert requests/second into requests per slot (multiply by slot length).
11. **CSMA/CD bit time:** $1/R$. 512 bit times at 100 Mbps = 5.12 µs.
12. Answer the "justify" part: the sample marks expect a one-line reason (e.g., "remainder ≠ 0 so error").

---

6. Textbook practice (Kurose & Ross, 9th ed.)

Topic	Problems
Delays, store-and-forward, pipelining	Ch. 1 P2, P5, P6, P7, P10, P11, P12, P13
Circuit vs packet switching	Ch. 1 P3, P4, P8, P9
Queueing delay, Little's law	Ch. 1 P13, P14, P15, P16
Throughput, bandwidth-delay	Ch. 1 P20, P21, P23, P25, P27, P28
Parity, CRC	Ch. 6 P1, P2, P5, P6, P7
ALOHA	Ch. 6 P8, P9, P10, P11
CSMA/CD	Ch. 6 P17, P18, P19, review R4, R6
Stop-and-wait / GBN / SR (Section 3.4.2 to 3.4.4)	Ch. 3 P15 (window for 98% utilization), P22 (GBN windows), P23 (SR window vs sequence space)

Quick answers for self-checking: Ch. 6 P5 ($D = 1010101010$, $G = 10011$) gives $R = 0100$. Ch. 6 P8: $p^* = 1/N$, efficiency $\rightarrow 1/e$. Ch. 6 R6: after the 5th collision $P(K = 4) = 1/32$; $K = 4$ at 10 Mbps is $4 \times 512 \times 0.1\ \mu s = 204.8\ \mu$s.

---

7. Concept self-test (cover the answers)

1. **Packet vs circuit switching: which allows more users and why?** Packet switching, through statistical multiplexing: bursty users rarely all transmit at once, so capacity is shared on demand.
2. **What is "circuit-like" behavior?** Guaranteed performance (constant rate, constant delay) from resources reserved along the path.
3. **Name the four delay components.** Processing, queueing, transmission, propagation.
4. **What happens as traffic intensity approaches 1?** Average queueing delay grows without bound.

5. **What does “M” mean in M/M/1?** Markovian (memoryless): Poisson arrivals and exponential service times.
6. **State Little’s law.** $E[N] = \lambda W$ (average number = arrival rate × average time in system).
7. **Why does statistical multiplexing give lower delay than TDM at the same load?** Scaling λ and μ by k keeps L fixed but divides W by k (Example 2.2).
8. **What error patterns does single parity miss?** Any even number of bit errors.
9. **What can 2D parity do that single parity cannot?** Locate and correct any single-bit error.
10. **Which bursts does CRC always detect?** All bursts of length $\leq r$ (the degree of G).
11. **Why does stop-and-wait need sequence numbers?** To detect duplicate frames when an ACK is lost or a timeout is premature (1 bit suffices).
12. **Go-Back-N vs Selective Repeat?** GBN receiver discards out-of-order frames and the sender resends the whole window; SR buffers out-of-order frames and resends only the missing ones.
13. **Three classes of MAC protocols?** Channel partitioning, random access, taking turns.
14. **Why is slotted ALOHA twice as efficient as pure ALOHA?** Vulnerable period drops from $2X$ to X.
15. **Why can CSMA still have collisions?** Propagation delay: a node may not yet hear a transmission that has already started.
16. **How long until CSMA/CD detects a collision (worst case)?** $2t_{prop}$.
17. **What problem do RTS/CTS solve?** The hidden terminal problem (and they make collisions cheap).
18. **Why does WiFi use link-layer ACKs but wired Ethernet does not?** Wireless links have high error rates and a sender cannot detect collisions at the receiver (hidden terminals).
19. **Purpose of bit stuffing?** Prevent the flag pattern `01111110` from appearing inside the frame data.
20. **Which layer does each device work at?** Switch: link layer (2). Router: network layer (3). Host: all five.